

Requirement Ambiguity Verification Report

Summary

Total Requirements	405
Ambiguous Requirements	25
Clear Requirements	380
Ambiguity Percentage	6.17%
Overall SRS Quality Score	98.72/100

Detailed Requirement Analysis

Requirement	Status	Severity	Score	Explanation
Copyright © 2002 by Karl E. Wiegers. Permission is granted to use, modify, and distribute this document.	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirements Specification	Clear	None	100%	No vague terms or weak verbs detected.
for	Clear	None	100%	No vague terms or weak verbs detected.
Requirement Ambiguity Detection and	Clear	None	100%	No vague terms or weak verbs detected.
Verification System	Clear	None	100%	No vague terms or weak verbs detected.
Version 1.0 approved	Clear	None	100%	No vague terms or weak verbs detected.
Prepared by	Clear	None	100%	No vague terms or weak verbs detected.
Adithyan P R: 23MID0035	Clear	None	100%	No vague terms or weak verbs detected.
Ramakrishnan R : 24MID0349	Clear	None	100%	No vague terms or weak verbs detected.
Hariprem K: 23MID0430	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirements Specification for Requirement Ambiguity Detection and Verification System Page ii	Clear	None	100%	No vague terms or weak verbs detected.
Table of Contents	Clear	None	100%	No vague terms or weak verbs detected.
Table of Contents ii	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
Revision History ii	Clear	None	100%	No vague terms or weak verbs detected.
1. Introduction 1	Clear	None	100%	No vague terms or weak verbs detected.
1.1 Purpose 1	Clear	None	100%	No vague terms or weak verbs detected.
1.2 Document Conventions 1	Clear	None	100%	No vague terms or weak verbs detected.
1.3 Intended Audience and Reading Suggestions 1	Clear	None	100%	No vague terms or weak verbs detected.
1.4 Project Scope 1	Clear	None	100%	No vague terms or weak verbs detected.
1.5 References 2	Clear	None	100%	No vague terms or weak verbs detected.
2. Overall Description 3	Clear	None	100%	No vague terms or weak verbs detected.
2.1 Product Perspective 3	Clear	None	100%	No vague terms or weak verbs detected.
2.2 Product Features 3	Clear	None	100%	No vague terms or weak verbs detected.
2.3 User Classes and Characteristics 3	Clear	None	100%	No vague terms or weak verbs detected.
2.4 Operating Environment 3	Clear	None	100%	No vague terms or weak verbs detected.
2.5 Design and Implementation Constraints 3	Clear	None	100%	No vague terms or weak verbs detected.
2.6 User Documentation 4	Clear	None	100%	No vague terms or weak verbs detected.
2.7 Assumptions and Dependencies 4	Clear	None	100%	No vague terms or weak verbs detected.
3. System Features (Functional Requirements) 4	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
3.1 System Feature 1 4	Clear	None	100%	No vague terms or weak verbs detected.
3.2 System Feature 2 5	Clear	None	100%	No vague terms or weak verbs detected.
3.3 System Feature 3 5	Clear	None	100%	No vague terms or weak verbs detected.
4. UML Diagrams 6	Clear	None	100%	No vague terms or weak verbs detected.
3.1 Use Case Diagram 6	Clear	None	100%	No vague terms or weak verbs detected.
3.2 Activity Diagram 7	Clear	None	100%	No vague terms or weak verbs detected.
3.3 Class Diagram 8	Clear	None	100%	No vague terms or weak verbs detected.
3.4 Sequence Diagram 9	Clear	None	100%	No vague terms or weak verbs detected.
5. External Interface Requirements 10	Clear	None	100%	No vague terms or weak verbs detected.
4.1 User Interfaces 10	Clear	None	100%	No vague terms or weak verbs detected.
4.2 Hardware Interfaces 11	Clear	None	100%	No vague terms or weak verbs detected.
4.3 Software Interfaces 12	Clear	None	100%	No vague terms or weak verbs detected.
4.4 Communications Interfaces 13	Clear	None	100%	No vague terms or weak verbs detected.
6. Other Nonfunctional Requirements 10	Clear	None	100%	No vague terms or weak verbs detected.
5.1 Performance Requirements 10	Clear	None	100%	No vague terms or weak verbs detected.
5.2 Safety Requirements 10	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
5.3 Security Requirements 11	Clear	None	100%	No vague terms or weak verbs detected.
5.4 Software Quality Attributes 11	Clear	None	100%	No vague terms or weak verbs detected.
7. Other Requirements 11	Clear	None	100%	No vague terms or weak verbs detected.
Appendix A: Glossary 11	Clear	None	100%	No vague terms or weak verbs detected.
Appendix B: Analysis Models11	Clear	None	100%	No vague terms or weak verbs detected.
Appendix C: Issues List 12	Clear	None	100%	No vague terms or weak verbs detected.
Revision History	Clear	None	100%	No vague terms or weak verbs detected.
Name Date Reason For Changes Version	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirements Specification for Requirement Ambiguity Detection and Verification System	Clear	None	100%	No vague terms or weak verbs detected.
Page 1`	Clear	None	100%	No vague terms or weak verbs detected.
1. Introduction	Clear	None	100%	No vague terms or weak verbs detected.
1.1 Purpose	Clear	None	100%	No vague terms or weak verbs detected.
The purpose of this Software Requirements Specification (SRS) document is to clearly describe	Clear	None	100%	No vague terms or weak verbs detected.
the requirements of the Requirement Ambiguity Detection and Verification System . This	Clear	None	100%	No vague terms or weak verbs detected.
document defines the system's functional and non -functional requirements, operating constraints,	Clear	None	100%	No vague terms or weak verbs detected.
external interfaces, and verification approach. The SRS acts as a formal agreement between	Clear	None	100%	No vague terms or weak verbs detected.
stakeholders and the development team and serves as a reference throughout the software	Clear	None	100%	No vague terms or weak verbs detected.
development life cycle. It ensures that all stakeholders have a common understanding of what the	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
system is expected to perform.	Clear	None	100%	No vague terms or weak verbs detected.
1.2 Document Conventions	Clear	None	100%	No vague terms or weak verbs detected.
This document follows standard SRS conventions. The word "shall" is used to indicate mandatory	Clear	None	100%	No vague terms or weak verbs detected.
requirements, while "should" represents optional features. Each requirement is uniquely identified	Ambiguous	Low	80%	Weak verb: should
for traceability. Tables and lists are used wherever clarity is required.	Clear	None	100%	No vague terms or weak verbs detected.
1.3 Intended Audience and Reading Suggestions	Clear	None	100%	No vague terms or weak verbs detected.
This document is intended for faculty members, evaluators, student developers, testers, and	Clear	None	100%	No vague terms or weak verbs detected.
reviewers involved in the project. Faculty and evaluators can use this document to assess the	Ambiguous	Low	80%	Weak verb: can
scope and completeness of the project, while developers and testers can use it as a guide during	Ambiguous	Low	80%	Weak verb: can
implementation and testing.	Clear	None	100%	No vague terms or weak verbs detected.
Readers are advised to begin with the Introduction and Overall Description sections to	Clear	None	100%	No vague terms or weak verbs detected.
understand the project background and objectives. The System Features section should then be	Ambiguous	Low	80%	Weak verb: should
referred to for detailed functional requirements.	Clear	None	100%	No vague terms or weak verbs detected.
1.4 Project Scope	Clear	None	100%	No vague terms or weak verbs detected.
The project focuses on developing a software system that verifies Software Requirement	Clear	None	100%	No vague terms or weak verbs detected.
Specification (SRS) documents before the implementation phase of software development. The	Clear	None	100%	No vague terms or weak verbs detected.
system analyses requirement statements to detect ambiguous, unclear, or loosely defined terms	Clear	None	100%	No vague terms or weak verbs detected.
that may lead to misunderstanding among stakeholders.	Ambiguous	Low	80%	Weak verb: may

Requirement	Status	Severity	Score	Explanation
The primary goal of this project is to support early -stage software verification and validation by	Clear	None	100%	No vague terms or weak verbs detected.
improving requirement quality. By identifying potential requirement issues at an early stage, the	Clear	None	100%	No vague terms or weak verbs detected.
system helps reduce errors, rework, and development costs in later phases of the software	Clear	None	100%	No vague terms or weak verbs detected.
development life cycle. The system is intended to be used as a supportive tool by students, testers,	Clear	None	100%	No vague terms or weak verbs detected.
and faculty to ensure better clarity and correctness of software requirements.	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirements Specification for Requirement Ambiguity Detection and Verification System	Clear	None	100%	No vague terms or weak verbs detected.
Page 2`	Clear	None	100%	No vague terms or weak verbs detected.
1.5 References	Clear	None	100%	No vague terms or weak verbs detected.
• IEEE Std 830 -1998, IEEE Recommended Practice for Software Requirements	Clear	None	100%	No vague terms or weak verbs detected.
Specifications , IEEE Computer Society, 1998.	Clear	None	100%	No vague terms or weak verbs detected.
• IEEE Std 29148 -2018, ISO/IEC/IEEE International Standard – Systems and Software	Clear	None	100%	No vague terms or weak verbs detected.
Engineering — Life Cycle Processes — Requirements Engineering , IEEE, 2018.	Clear	None	100%	No vague terms or weak verbs detected.
• Sommerville, I., Software Engineering , 10th Edition, Pearson Education, 2016.	Clear	None	100%	No vague terms or weak verbs detected.
• Pressman, R. S., and Maxim, B. R., Software Engineering: A Practitioner's Approach , 8th	Clear	None	100%	No vague terms or weak verbs detected.
Edition, McGraw -Hill, 2015.	Clear	None	100%	No vague terms or weak verbs detected.
• Wiegers, K. E., and Beatty, J., Software Requirements , 3rd Edition, Microsoft Press, 2013.	Clear	None	100%	No vague terms or weak verbs detected.
• ISO/IEC 25010:2011, Systems and Software Engineering — Systems and Software Quality	Clear	None	100%	No vague terms or weak verbs detected.
Requirements and Evaluation (SQuaRE) .	Clear	None	100%	No vague terms or weak verbs detected.
• Boehm, B., Software Engineering Economics , Prentice Hall, 1981.	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
• Myers, G. J., Sandler, C., and Badgett, T., The Art of Software Testing , 3rd Edition, Wiley,	Clear	None	100%	No vague terms or weak verbs detected.
2011.	Clear	None	100%	No vague terms or weak verbs detected.
• IEEE Std 1012 -2016, IEEE Standard for System, Software, and Hardware Verification and	Clear	None	100%	No vague terms or weak verbs detected.
Validation .	Clear	None	100%	No vague terms or weak verbs detected.
• IEEE Std 1028 -2008, IEEE Standard for Software Reviews and Audits .	Clear	None	100%	No vague terms or weak verbs detected.
• Beck, K. et al., Manifesto for Agile Software Development , 2001.	Clear	None	100%	No vague terms or weak verbs detected.
https://agilemanifesto.org	Clear	None	100%	No vague terms or weak verbs detected.
• Schwaber, K., and Sutherland, J., The Scrum Guide , 2020.	Clear	None	100%	No vague terms or weak verbs detected.
https://scrumguides.org	Clear	None	100%	No vague terms or weak verbs detected.
• Leffingwell, D., Agile Software Requirements , Addison -Wesley, 2011.	Clear	None	100%	No vague terms or weak verbs detected.
• Juristo, N., and Moreno, A. M., Basics of Software Engineering Experimentation , Springer,	Clear	None	100%	No vague terms or weak verbs detected.
2001.	Clear	None	100%	No vague terms or weak verbs detected.
• Kitchenham, B., Pfleeger, S. L., Software Quality: The Elusive Target , IEEE Software,	Clear	None	100%	No vague terms or weak verbs detected.
1996.	Clear	None	100%	No vague terms or weak verbs detected.
• Nuseibeh, B., and Easterbrook, S., Requirements Engineering: A Roadmap , Proceedings of	Clear	None	100%	No vague terms or weak verbs detected.
the Conference on the Future of Software Engineering, 2000.	Clear	None	100%	No vague terms or weak verbs detected.
• Pohl, K., Requirements Engineering: Fundamentals, Principles, and Techniques , Springer,	Clear	None	100%	No vague terms or weak verbs detected.
2010.	Clear	None	100%	No vague terms or weak verbs detected.
• Firesmith, D., Common Requirements Problems, Their Negative Consequences, and the	Clear	None	100%	No vague terms or weak verbs detected.
Industry Best Practices to Help Solve Them , Journal of Object Technology, 2007.	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
• Glinz, M., On Non -Functional Requirements , IEEE Requirements Engineering Conference,	Clear	None	100%	No vague terms or weak verbs detected.
2007.	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirements Specification for Requirement Ambiguity Detection and Verification System	Clear	None	100%	No vague terms or weak verbs detected.
Page 3`	Clear	None	100%	No vague terms or weak verbs detected.
• Process Impact, Software Requirements Specification Templates and Guidelines .	Clear	None	100%	No vague terms or weak verbs detected.
http://www.processimpact.com	Clear	None	100%	No vague terms or weak verbs detected.
2. Overall Description	Clear	None	100%	No vague terms or weak verbs detected.
2.1 Product Perspective	Clear	None	100%	No vague terms or weak verbs detected.
The Requirement Ambiguity Detection and Verification System is a standalone, web -based	Clear	None	100%	No vague terms or weak verbs detected.
software application developed to support the requirement analysis phase of software	Clear	None	100%	No vague terms or weak verbs detected.
development. The system is not intended to replace existing requirement management tools but	Clear	None	100%	No vague terms or weak verbs detected.
acts as an assisting tool that verifies the quality of requirement documents before implementation	Clear	None	100%	No vague terms or weak verbs detected.
begins. It operates independently and processes Software Requirement Specification documents	Clear	None	100%	No vague terms or weak verbs detected.
uploaded by users. The system fits into the early stage of the software development life cycle and	Clear	None	100%	No vague terms or weak verbs detected.
helps improve communication between stakeholders by ensuring requirement clarity.	Clear	None	100%	No vague terms or weak verbs detected.
2.2 User Classes and Characteristics	Clear	None	100%	No vague terms or weak verbs detected.
The system is intended to be used by students, faculty members, and testers involved in software	Clear	None	100%	No vague terms or weak verbs detected.
development and evaluation activities. Students use the system to verify requirement documents	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
prepared as part of academic projects. Faculty members and evaluators use the system to review	Clear	None	100%	No vague terms or weak verbs detected.
verification results and assess requirement quality. Users are expected to have basic knowledge of	Clear	None	100%	No vague terms or weak verbs detected.
computer usage and web -based applications. No advanced technical skills are required to operate	Clear	None	100%	No vague terms or weak verbs detected.
the system.	Clear	None	100%	No vague terms or weak verbs detected.
2.3 Operating Environment	Clear	None	100%	No vague terms or weak verbs detected.
The system is designed to operate on standard computing platforms running Windows or Linux	Clear	None	100%	No vague terms or weak verbs detected.
operating systems. Users interact with the system through commonly used web browsers such as	Clear	None	100%	No vague terms or weak verbs detected.
Google Chrome or Mozilla Firefox. The user interface of the system is developed using HTML and	Clear	None	100%	No vague terms or weak verbs detected.
CSS, while the core processing logic is implemented using Python. Open -source text processing	Clear	None	100%	No vague terms or weak verbs detected.
libraries are used to analyse Software Requirement Specification documents.	Clear	None	100%	No vague terms or weak verbs detected.
Selenium is used as a software testing and verification tool for this project. It supports	Clear	None	100%	No vague terms or weak verbs detected.
automated testing of the web -based application by simulating user interactions such as document	Clear	None	100%	No vague terms or weak verbs detected.
upload and report viewing. Selenium helps verify that the system functions correctly under different	Clear	None	100%	No vague terms or weak verbs detected.
scenarios and ensures that functional requirements are met. By using Selenium, the project	Clear	None	100%	No vague terms or weak verbs detected.
supports systematic verification and validation of system behaviour without manual intervention.	Clear	None	100%	No vague terms or weak verbs detected.
2.4 Design and Implementation Constraints	Clear	None	100%	No vague terms or weak verbs detected.
The development of the system is constrained by the requirement to use open -source tools and	Clear	None	100%	No vague terms or weak verbs detected.
technologies suitable for academic purposes. The system must be lightweight and capable of	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
running on standard hardware without requiring specialized infrastructure. Design simplicity and	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirements Specification for Requirement Ambiguity Detection and Verification System	Clear	None	100%	No vague terms or weak verbs detected.
Page 4`	Clear	None	100%	No vague terms or weak verbs detected.
maintainability are prioritized to ensure ease of development and future enhancement. The system	Clear	None	100%	No vague terms or weak verbs detected.
must comply with standard software engineering practices and documentation guidelines.	Clear	None	100%	No vague terms or weak verbs detected.
2.5 User Documentation	Clear	None	100%	No vague terms or weak verbs detected.
Basic user documentation will be provided to assist users in understanding how to operate the	Clear	None	100%	No vague terms or weak verbs detected.
system. This documentation will include instructions for uploading requirement documents and	Clear	None	100%	No vague terms or weak verbs detected.
interpreting the generated verification reports. The documentation is intended to be simple and	Clear	None	100%	No vague terms or weak verbs detected.
accessible to users with minimal technical background.	Clear	None	100%	No vague terms or weak verbs detected.
2.6 Assumptions and Dependencies	Clear	None	100%	No vague terms or weak verbs detected.
It is assumed that users will upload properly formatted Software Requirement Specification (SRS)	Clear	None	100%	No vague terms or weak verbs detected.
documents. The effectiveness of ambiguity detection depends on the clarity and structure of the	Clear	None	100%	No vague terms or weak verbs detected.
input document. The system assumes the availability of a standard operating environment with a	Clear	None	100%	No vague terms or weak verbs detected.
supported web browser.	Clear	None	100%	No vague terms or weak verbs detected.
The project depends on open -source text processing libraries and testing tools used for verification	Clear	None	100%	No vague terms or weak verbs detected.
and validation. Any changes in these tools or development schedule may affect system	Ambiguous	Low	80%	Weak verb: may
functionality.	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
3. System Features (Functional Requirements)	Clear	None	100%	No vague terms or weak verbs detected.
This section describes the major functional features provided by the Requirement Ambiguity	Clear	None	100%	No vague terms or weak verbs detected.
Detection and Verification System. Each feature explains how the system responds to user actions	Clear	None	100%	No vague terms or weak verbs detected.
and the functionality it delivers to support requirement verification.	Clear	None	100%	No vague terms or weak verbs detected.
3.1 System Feature 1 : SRS Document Upload	Clear	None	100%	No vague terms or weak verbs detected.
3.1.1 Description and Priority	Clear	None	100%	No vague terms or weak verbs detected.
This feature enables users to upload Software Requirement Specification documents	Clear	None	100%	No vague terms or weak verbs detected.
for analysis. The system provides an interface through which users can select and	Ambiguous	Low	80%	Weak verb: can
submit their required documents. Once uploaded, the system verifies whether the	Clear	None	100%	No vague terms or weak verbs detected.
document is in a supported format before processing. This feature ensures that only	Clear	None	100%	No vague terms or weak verbs detected.
valid input documents are accepted for analysis, thereby reducing errors during	Clear	None	100%	No vague terms or weak verbs detected.
further processing. Since document upload is the first step in the verification process,	Clear	None	100%	No vague terms or weak verbs detected.
this feature is considered a high -priority functionality.	Clear	None	100%	No vague terms or weak verbs detected.
3.1.2 Stimulus/Response Sequences	Clear	None	100%	No vague terms or weak verbs detected.
• User selects an SRS document	Clear	None	100%	No vague terms or weak verbs detected.
• System validates the document format	Clear	None	100%	No vague terms or weak verbs detected.
• System accepts the document for processing	Clear	None	100%	No vague terms or weak verbs detected.
3.1.3 Functional Requirements	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirements Specification for Requirement Ambiguity Detection and Verification System	Clear	None	100%	No vague terms or weak verbs detected.
Page 5`	Clear	None	100%	No vague terms or weak verbs detected.
REQ -1: The system shall allow users to upload SRS documents.	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
REQ -2: The system shall validate the uploaded document format.	Clear	None	100%	No vague terms or weak verbs detected.
3.2 System Feature 2 : Requirement Extraction and Ambiguity Detection	Clear	None	100%	No vague terms or weak verbs detected.
3.2.1 Description and Priority	Clear	None	100%	No vague terms or weak verbs detected.
This feature is responsible for analyzing the uploaded SRS document and identifying	Clear	None	100%	No vague terms or weak verbs detected.
ambiguous requirement statements. After the document is accepted, the system	Clear	None	100%	No vague terms or weak verbs detected.
extracts individual requirement statements from the content. Each statement is then	Clear	None	100%	No vague terms or weak verbs detected.
evaluated to detect ambiguous or unclear terms that may lead to misinterpretation	Ambiguous	Low	80%	Weak verb: may
during software development. The system focuses on commonly used vague words	Clear	None	100%	No vague terms or weak verbs detected.
and expressions that reduce requirement clarity. Detected ambiguous statements are	Clear	None	100%	No vague terms or weak verbs detected.
flagged for further review. This feature plays a critical role in early -stage software	Clear	None	100%	No vague terms or weak verbs detected.
verification and is treated as a high -priority function.	Clear	None	100%	No vague terms or weak verbs detected.
3.2.2 Stimulus/Response Sequences	Clear	None	100%	No vague terms or weak verbs detected.
• System scans requirement statement	Ambiguous	Low	80%	Weak verb: can
• Ambiguous terms are identified	Clear	None	100%	No vague terms or weak verbs detected.
• Problematic statements are marked	Clear	None	100%	No vague terms or weak verbs detected.
3.2.3 Functional Requirements	Clear	None	100%	No vague terms or weak verbs detected.
REQ -3: The system shall extract individual requirement statements.	Clear	None	100%	No vague terms or weak verbs detected.
REQ -4: The system shall detect ambiguous words such as "fast", "user-friendly",	Ambiguous	Low	80%	Vague term: fast
and "adequate".	Clear	None	100%	No vague terms or weak verbs detected.
3.3 System Feature 3: Inconsistency Identification	Clear	None	100%	No vague terms or weak verbs detected.
3.3.1 Description and Priority	Clear	None	100%	No vague terms or weak verbs detected.
In this feature, the system checks for basic inconsistencies within the requirement	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
document. The system analyzes terminology usage across multiple requirement	Clear	None	100%	No vague terms or weak verbs detected.
statements to identify variations or conflicts in naming or description. Inconsistent	Clear	None	100%	No vague terms or weak verbs detected.
usage of terms can create confusion among stakeholders and developers. By	Ambiguous	Low	80%	Weak verb: can
identifying such issues, the system supports improved requirement consistency and	Clear	None	100%	No vague terms or weak verbs detected.
understanding. This feature enhances the overall quality of the requirement	Clear	None	100%	No vague terms or weak verbs detected.
document and complements ambiguity detection.	Clear	None	100%	No vague terms or weak verbs detected.
3.3.2 Stimulus/Response Sequences	Clear	None	100%	No vague terms or weak verbs detected.
• System compiles detected issues	Clear	None	100%	No vague terms or weak verbs detected.
• Verification report is generated	Clear	None	100%	No vague terms or weak verbs detected.
• User views or downloads the report	Clear	None	100%	No vague terms or weak verbs detected.
3.3.3 Functional Requirements	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirements Specification for Requirement Ambiguity Detection and Verification System	Clear	None	100%	No vague terms or weak verbs detected.
Page 6`	Clear	None	100%	No vague terms or weak verbs detected.
REQ -5: The system shall generate a verification report.	Clear	None	100%	No vague terms or weak verbs detected.
REQ -6: The system shall allow users to view the report on screen.	Clear	None	100%	No vague terms or weak verbs detected.
4. UML Diagrams	Clear	None	100%	No vague terms or weak verbs detected.
4.1 Use Case Diagram	Clear	None	100%	No vague terms or weak verbs detected.
The Use Case Diagram represents the overall functionality of the Requirement Ambiguity Detection	Clear	None	100%	No vague terms or weak verbs detected.
System and illustrates how the user interacts with the system. The system boundary encloses all	Clear	None	100%	No vague terms or weak verbs detected.
the functionalities provided by the software, while the external actor is the User, who may be a	Ambiguous	Low	80%	Weak verb: may

Requirement	Status	Severity	Score	Explanation
student, faculty member, or tester.	Clear	None	100%	No vague terms or weak verbs detected.
The process begins when the user uploads an SRS document to the system. This is the primary	Clear	None	100%	No vague terms or weak verbs detected.
interaction and serves as the input to the system. Once the document is uploaded, the system	Clear	None	100%	No vague terms or weak verbs detected.
extracts the text content from the file so that it can be processed further. After text extraction, the	Ambiguous	Low	80%	Weak verb: can
system analyses the requirements present in the document. During this analysis phase, the system	Clear	None	100%	No vague terms or weak verbs detected.
identifies ambiguous words and unclear statements that may lead to misinterpretation during	Ambiguous	Low	80%	Weak verb: may
software development.	Clear	None	100%	No vague terms or weak verbs detected.
Following the analysis, the system generates a verification report that summarizes all detected	Clear	None	100%	No vague terms or weak verbs detected.
issues. In addition to listing ambiguous requirements, the system calculates a quality score to	Clear	None	100%	No vague terms or weak verbs detected.
indicate the overall clarity and quality of the requirement document. The system also classifies the	Clear	None	100%	No vague terms or weak verbs detected.
severity of detected issues, helping users understand whether the ambiguity is minor, moderate, or	Clear	None	100%	No vague terms or weak verbs detected.
critical.	Clear	None	100%	No vague terms or weak verbs detected.
Once the analysis is complete, the user can view the results through the verification dashboard,	Ambiguous	Low	80%	Weak verb: can
where all findings, scores, and classifications are displayed in a structured format. Finally, the	Clear	None	100%	No vague terms or weak verbs detected.
system provides an option for the user to download the verification report as a PDF document for	Clear	None	100%	No vague terms or weak verbs detected.
future reference or evaluation purposes.	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirements Specification for Requirement Ambiguity Detection and Verification System	Clear	None	100%	No vague terms or weak verbs detected.
Page 7`	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
Overall, the use case diagram demonstrates the complete workflow of the system, starting from	Clear	None	100%	No vague terms or weak verbs detected.
document upload and ending with report generation and download, clearly showing how the	Clear	None	100%	No vague terms or weak verbs detected.
system supports requirement verification and ambiguity detection.	Clear	None	100%	No vague terms or weak verbs detected.
4.2 Activity Diagram	Clear	None	100%	No vague terms or weak verbs detected.
The activity diagram shows the step-by-step workflow of the Requirement Ambiguity Detection	Clear	None	100%	No vague terms or weak verbs detected.
System. The process starts when the user uploads an SRS document. The system then extracts	Clear	None	100%	No vague terms or weak verbs detected.
text from the file and splits it into individual requirement statements.	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirements Specification for Requirement Ambiguity Detection and Verification System	Clear	None	100%	No vague terms or weak verbs detected.
Page 8`	Clear	None	100%	No vague terms or weak verbs detected.
Next, the system analyses each requirement by detecting vague words and weak verbs. After this	Clear	None	100%	No vague terms or weak verbs detected.
analysis, a decision is made to check whether any ambiguity is found. If ambiguity is detected, the	Clear	None	100%	No vague terms or weak verbs detected.
system assigns a severity level and a score to that requirement. If no ambiguity is found, the	Clear	None	100%	No vague terms or weak verbs detected.
requirement is marked as clear.	Clear	None	100%	No vague terms or weak verbs detected.
After all requirements are evaluated, the system calculates the overall SRS quality score. The	Clear	None	100%	No vague terms or weak verbs detected.
results are then displayed on the dashboard, and a PDF verification report is generated. The	Clear	None	100%	No vague terms or weak verbs detected.
process ends after the report is created.	Clear	None	100%	No vague terms or weak verbs detected.
4.3 Class Diagram	Clear	None	100%	No vague terms or weak verbs detected.
The class diagram represents the structural design of the Requirement Ambiguity Detection	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
System. It shows the main classes in the system and the relationships between them.	Clear	None	100%	No vague terms or weak verbs detected.
The central class in the system is the WebController. This class acts as the main control unit that	Clear	None	100%	No vague terms or weak verbs detected.
handles user interactions. It contains methods such as uploadFile() and displayDashboard(). When	Clear	None	100%	No vague terms or weak verbs detected.
a user uploads an SRS document, the WebController manages the request and coordinates further	Clear	None	100%	No vague terms or weak verbs detected.
processing.	Clear	None	100%	No vague terms or weak verbs detected.
The WebController interacts with the FileHandler class. The FileHandler is responsible for handling	Clear	None	100%	No vague terms or weak verbs detected.
file-related operations. It contains the method extractText(), which extracts text content from the	Clear	None	100%	No vague terms or weak verbs detected.
uploaded document so that it can be analyzed.	Ambiguous	Low	80%	Weak verb: can
After text extraction, the WebController communicates with the AmbiguityAnalyzer class. This class	Clear	None	100%	No vague terms or weak verbs detected.
performs the core processing of the system. It contains methods such as detectVagueWords(),	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirements Specification for Requirement Ambiguity Detection and Verification System	Clear	None	100%	No vague terms or weak verbs detected.
Page 9`	Clear	None	100%	No vague terms or weak verbs detected.
detectWeakVerbs(), classifySeverity(), and calculateScore(). These methods analyze each	Clear	None	100%	No vague terms or weak verbs detected.
requirement statement, detect ambiguity, assign severity levels, and calculate quality scores.	Clear	None	100%	No vague terms or weak verbs detected.
Once the analysis is complete, the system interacts with the ReportGenerator class. This class	Clear	None	100%	No vague terms or weak verbs detected.
contains the method generatePDF(), which creates the final verification report in PDF format.	Clear	None	100%	No vague terms or weak verbs detected.
4.4 Sequence Diagram	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
The sequence diagram illustrates the interaction between the User, Web Interface, Ambiguity	Clear	None	100%	No vague terms or weak verbs detected.
Engine, and PDF Generator during the execution of the Requirement Ambiguity Detection System.	Clear	None	100%	No vague terms or weak verbs detected.
The process begins when the user uploads an SRS file through the Web Interface. After receiving	Clear	None	100%	No vague terms or weak verbs detected.
the file, the Web Interface extracts the text and sends the extracted content to the Ambiguity	Clear	None	100%	No vague terms or weak verbs detected.
Engine for analysis.	Clear	None	100%	No vague terms or weak verbs detected.
The Ambiguity Engine processes the requirements by detecting vague words, weak verbs, and	Clear	None	100%	No vague terms or weak verbs detected.
other ambiguities. Once the analysis is completed, it returns the analysis results back to the Web	Clear	None	100%	No vague terms or weak verbs detected.
Interface.	Clear	None	100%	No vague terms or weak verbs detected.
The Web Interface then displays the verification dashboard to the user, showing the analysis	Clear	None	100%	No vague terms or weak verbs detected.
results, quality score, and detected issues.	Clear	None	100%	No vague terms or weak verbs detected.
If the user chooses to download the report, the Web Interface sends a request to the PDF	Clear	None	100%	No vague terms or weak verbs detected.
Generator to generate the report. The PDF Generator creates the PDF document and sends the	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirements Specification for Requirement Ambiguity Detection and Verification System	Clear	None	100%	No vague terms or weak verbs detected.
Page 10`	Clear	None	100%	No vague terms or weak verbs detected.
generated file back to the Web Interface. Finally, the Web Interface delivers the PDF file to the	Clear	None	100%	No vague terms or weak verbs detected.
user for download.	Clear	None	100%	No vague terms or weak verbs detected.
5. External Interface Requirements	Clear	None	100%	No vague terms or weak verbs detected.
This section describes the interfaces through which the system interacts with users, hardware,	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
software components, and communication mechanisms. These interfaces define how the system	Clear	None	100%	No vague terms or weak verbs detected.
exchanges information with external entities.	Clear	None	100%	No vague terms or weak verbs detected.
5.1 User Interfaces	Clear	None	100%	No vague terms or weak verbs detected.
The system provides a simple and user -friendly web -based interface that allows users to interact	Clear	None	100%	No vague terms or weak verbs detected.
with the application through a standard web browser. The user interface enables users to upload	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirement Specification documents and view the generated verification reports. The	Clear	None	100%	No vague terms or weak verbs detected.
interface is designed to be intuitive, with clear labels and instructions, so that users with basic	Clear	None	100%	No vague terms or weak verbs detected.
computer knowledge can easily operate the system. Error messages and system notifications are	Ambiguous	Low	80%	Weak verb: can
displayed in a clear manner to guide users during interaction.	Clear	None	100%	No vague terms or weak verbs detected.
5.2 Hardware Interfaces	Clear	None	100%	No vague terms or weak verbs detected.
The system does not require any specialized hardware components. It operates on standard	Clear	None	100%	No vague terms or weak verbs detected.
computing devices such as personal computers or laptops. All processing is performed using the	Clear	None	100%	No vague terms or weak verbs detected.
existing hardware resources of the system on which the application is deployed. No direct	Clear	None	100%	No vague terms or weak verbs detected.
interaction with external hardware devices is required.	Clear	None	100%	No vague terms or weak verbs detected.
5.3 Software Interfaces	Clear	None	100%	No vague terms or weak verbs detected.
The system interacts with various software components to perform its functions. It uses Python -	Clear	None	100%	No vague terms or weak verbs detected.
based text processing libraries to analyze requirement documents and detect ambiguous or	Clear	None	100%	No vague terms or weak verbs detected.
inconsistent statements. The system also interfaces with automated testing tools to support	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
verification and validation activities. These software interfaces operate internally and are	Clear	None	100%	No vague terms or weak verbs detected.
transparent to the end user, ensuring smooth system functionality.	Clear	None	100%	No vague terms or weak verbs detected.
5.4 Communications Interfaces	Clear	None	100%	No vague terms or weak verbs detected.
The system uses standard communication protocols to enable interaction between the client and	Clear	None	100%	No vague terms or weak verbs detected.
server components. Communication is carried out over HTTP using a web browser. The system	Clear	None	100%	No vague terms or weak verbs detected.
does not require any additional communication interfaces such as email or external messaging	Clear	None	100%	No vague terms or weak verbs detected.
services. Data exchanged during communication is limited to document content and verification	Clear	None	100%	No vague terms or weak verbs detected.
results.	Clear	None	100%	No vague terms or weak verbs detected.
6. Other Nonfunctional Requirements	Clear	None	100%	No vague terms or weak verbs detected.
This section describes the non -functional requirements that define the quality attributes and	Clear	None	100%	No vague terms or weak verbs detected.
constraints of the Requirement Ambiguity Detection and Verification System. These requirements	Clear	None	100%	No vague terms or weak verbs detected.
ensure that the system performs efficiently, securely, and reliably.	Ambiguous	Medium	60%	Vague term: efficient; Vague term: secure
Software Requirements Specification for Requirement Ambiguity Detection and Verification System	Clear	None	100%	No vague terms or weak verbs detected.
Page 11`	Clear	None	100%	No vague terms or weak verbs detected.
6.1 Performance Requirements	Clear	None	100%	No vague terms or weak verbs detected.
The system shall provide acceptable performance under normal operating conditions. It is	Clear	None	100%	No vague terms or weak verbs detected.
expected that the system will process and analyze uploaded Software Requirement Specification	Clear	None	100%	No vague terms or weak verbs detected.
documents within a reasonable amount of time. The performance should remain consistent for	Ambiguous	Low	80%	Weak verb: should

Requirement	Status	Severity	Score	Explanation
documents of moderate size, ensuring a smooth user experience without noticeable delays.	Clear	None	100%	No vague terms or weak verbs detected.
6.2 Safety Requirements	Clear	None	100%	No vague terms or weak verbs detected.
The system does not perform any safety -critical operations and does not interact with physical	Clear	None	100%	No vague terms or weak verbs detected.
hardware or real -time control systems. Therefore, there are no specific safety requirements	Clear	None	100%	No vague terms or weak verbs detected.
associated with this system. The system is intended solely for academic and verification purposes.	Clear	None	100%	No vague terms or weak verbs detected.
6.3 Security Requirements	Clear	None	100%	No vague terms or weak verbs detected.
The system shall ensure basic security measures to protect user data and uploaded documents.	Clear	None	100%	No vague terms or weak verbs detected.
Uploaded SRS documents should not be stored permanently on the system after analysis. Access	Ambiguous	Low	80%	Weak verb: should
to the system shall be limited to authorized users, and the system should prevent unauthorized	Ambiguous	Low	80%	Weak verb: should
access or misuse. These measures help maintain data confidentiality and user trust.	Clear	None	100%	No vague terms or weak verbs detected.
6.4 Software Quality Attributes	Clear	None	100%	No vague terms or weak verbs detected.
The system shall exhibit important software quality attributes such as reliability, usability, and	Clear	None	100%	No vague terms or weak verbs detected.
maintainability. The system should operate consistently without unexpected failures during normal	Ambiguous	Low	80%	Weak verb: should
use. It should be easy for users with minimal technical knowledge. The design should support ease	Ambiguous	Low	80%	Weak verb: should
of maintenance and future enhancements, allowing the system to be extended with additional	Clear	None	100%	No vague terms or weak verbs detected.
verification features if required.	Clear	None	100%	No vague terms or weak verbs detected.
7. Other Requirements	Clear	None	100%	No vague terms or weak verbs detected.
This section describes additional requirements that are not covered in previous sections but are	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
necessary for the successful development and management of the project.	Clear	None	100%	No vague terms or weak verbs detected.
The development of the Requirement Ambiguity Detection and Verification System will follow the	Clear	None	100%	No vague terms or weak verbs detected.
Scrum -based Agile development model , which supports iterative and incremental	Clear	None	100%	No vague terms or weak verbs detected.
development . The system will be developed through multiple short development cycles called	Clear	None	100%	No vague terms or weak verbs detected.
sprints. Each sprint represents one iteration of the development process and results in a working	Clear	None	100%	No vague terms or weak verbs detected.
increment of the system.	Clear	None	100%	No vague terms or weak verbs detected.
Iterative development allows the system to be built gradually, starting with basic functionality and	Clear	None	100%	No vague terms or weak verbs detected.
improving step by step in subsequent iterations. In the early iterations, core features such as	Clear	None	100%	No vague terms or weak verbs detected.
document upload and basic requirement extraction are implemented. Later iterations focus on	Clear	None	100%	No vague terms or weak verbs detected.
advanced functionalities such as ambiguity detection, inconsistency identification, and verification	Clear	None	100%	No vague terms or weak verbs detected.
report generation. This approach ensures that the system evolves continuously based on testing	Clear	None	100%	No vague terms or weak verbs detected.
results and feedback.	Clear	None	100%	No vague terms or weak verbs detected.
Software Requirements Specification for Requirement Ambiguity Detection and Verification System	Clear	None	100%	No vague terms or weak verbs detected.
Page 12`	Clear	None	100%	No vague terms or weak verbs detected.
Sprint -based planning is used to organize development activities in a structured manner. At the	Clear	None	100%	No vague terms or weak verbs detected.
end of each sprint, the implemented features are reviewed and verified against the requirements	Clear	None	100%	No vague terms or weak verbs detected.
defined in this SRS. This continuous review process helps in early identification of defects and	Clear	None	100%	No vague terms or weak verbs detected.
supports effective verification and validation throughout development.	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
The use of iterative development and Scrum is particularly suitable for this project because	Clear	None	100%	No vague terms or weak verbs detected.
requirement verification rules may change or expand during development. This approach improves	Ambiguous	Low	80%	Weak verb: may
flexibility, enhances team collaboration, and ensures that quality is maintained at every stage of the	Clear	None	100%	No vague terms or weak verbs detected.
project lifecycle.	Clear	None	100%	No vague terms or weak verbs detected.
Appendix A: Glossary	Clear	None	100%	No vague terms or weak verbs detected.
This glossary defines terms and abbreviations used in this document to ensure clarity and common	Clear	None	100%	No vague terms or weak verbs detected.
understanding.	Clear	None	100%	No vague terms or weak verbs detected.
• SRS (Software Requirements Specification): A document that describes the functional	Clear	None	100%	No vague terms or weak verbs detected.
and non -functional requirements of a software system.	Clear	None	100%	No vague terms or weak verbs detected.
• SVV (Software Verification and Validation): The process of evaluating software to ensure	Clear	None	100%	No vague terms or weak verbs detected.
it meets specified requirements and user needs.	Clear	None	100%	No vague terms or weak verbs detected.
• Requirement Ambiguity: The presence of unclear or vague statements in requirements	Clear	None	100%	No vague terms or weak verbs detected.
that may lead to multiple interpretations.	Ambiguous	Low	80%	Weak verb: may
• Verification: The process of checking whether the software meets specified requirements.	Clear	None	100%	No vague terms or weak verbs detected.
• Validation: The process of ensuring the software fulfils its intended purpose.	Clear	None	100%	No vague terms or weak verbs detected.
Appendix B: Analysis Models	Clear	None	100%	No vague terms or weak verbs detected.
Analysis models are used to represent the system behavior and structure during the design phase.	Clear	None	100%	No vague terms or weak verbs detected.
For this project, Unified Modeling Language (UML) diagrams such as Use Case Diagrams and	Clear	None	100%	No vague terms or weak verbs detected.

Requirement	Status	Severity	Score	Explanation
Activity Diagrams will be created to visually describe system interactions and workflow. These	Clear	None	100%	No vague terms or weak verbs detected.
models help in better understanding system functionality and support effective design and	Clear	None	100%	No vague terms or weak verbs detected.
implementation.	Clear	None	100%	No vague terms or weak verbs detected.
Appendix C: Issues List	Clear	None	100%	No vague terms or weak verbs detected.
At the time of preparation of this SRS document, there are no unresolved issues affecting the	Clear	None	100%	No vague terms or weak verbs detected.
system requirements. Any future issues, changes, or clarifications identified during development	Clear	None	100%	No vague terms or weak verbs detected.
will be documented and addressed through proper review and update of this document.	Clear	None	100%	No vague terms or weak verbs detected.